

Week 7 Weekly Report

Week 7 integrates the repaired Week 6 direct-price model into counterfactual stress tests and VIX-focused interpretation, retaining the market-feature updater and Streamlit prototype. VIX remains a volatility indicator, not a news-sentiment substitute.

1 Work Completed

- Regenerated VIX-related SHAP shares for the unchanged Random Forest volatility model and revised bounded direct-price model.
- Applied the required 50% volatility spike and absolute 2 percentage-point rate hike to the structural BSM model.
- Applied the same counterfactual inputs to both machine-learning pricing routes.
- Recorded the absence of a licensed or reproducible news-sentiment series instead of fabricating a feature.
- Built a JPM, Cboe VIX, FRED DGS10 and DGS1 updater with bounded retries, independent source caches, and complete feature reconstruction at a common completed date.
- Built and locally health-tested a Streamlit pricing and sensitivity prototype.

2 Deliverables

- Week7_Advanced_Analysis.ipynb
- sensitivity_analysis.py
- data_update.py
- Streamlit app.py
- tool_config.json
- Comprehensive Sensitivity Analysis Report
 - stress, SHAP, and data-status outputs

3 Progress Summary

Structural shocks, revised ML counterfactuals and sensitivity reporting are complete. Both applications retain automatic and manual refresh with tested fallback controls. The original research sample and Approach 1 are unchanged; Approach 2 and its downstream results are updated.

4 Experimental Results

Scenario	BSM	Approach 1	Approach 2
Base	99.2617	91.7072	91.8375
50% volatility spike	115.6683	92.0449	91.9144
2% rate hike	101.3317	94.5557	94.5690
Combined shock	116.9177	94.8010	94.5776

Figure 1 Structural chooser stress scenarios



5 Issues Encountered

- Public endpoints can time out or publish on different dates. Each request uses a 12-second timeout and at most two attempts. Per-source caches preserve available histories, and incomplete updates retain a labeled complete snapshot. FRED credentials stay outside logs and saved data.
- A news-sentiment feature requires a defined source, timestamp policy, and leakage controls. These inputs were not present in Weeks 1 and 2, so sentiment remains a documented data gap.
- ML counterfactuals can combine inputs that are uncommon in the training sample; results must be read with the training-domain limits.

6 Plan for Week 8

- Package BSM and the best ML model into a self-contained final tool.
- Display an empirical proxy-error margin and model limitations.
- Prepare the deployment checklist, final report, presentation, and demo script.